I/O Devices- Smart Rings

Smart rings are a type of input and output device that have become more popular recently. There are multiple brands that have grown in popularity recently, Oura Rings being the one that I've heard the most talk about recently. These smart rings are interesting because they're an incredibly small form factor, fitting on your finger, however they can have an impressive amount of capabilities. Many smart rings can connect to come other device and then respond to small motions such as tapping, or a small hand wave to signal a mouse clicking on something or moving. A lot of smart rings also have health monitoring features, similarly to how many smart watches do, by measuring heart rate, temperature, or sleep cycles. They can also be used as security devices similarly to how a key fob would work by tapping it onto an electronic lock and it grants authorization to certain things. A more common use than that, however, is using it for payments. Instead of getting out your phone or card for tap to pay, some smart rings will use NFC to allow payment to happen through them. They also have uses for output such as haptic feedback, where if a connected device receives a notification, they will buzz to alert the wearer. Many smart rings also have LED which can blink with codes to give feedback to the wearer. I think some interesting uses that we could see would be for accessibility for people with limited mobility. This could allow them to control devices without needing as much motion by just moving their hands slightly. Another thing these could be used for would be VR gaming. Instead of the current types of controllers, which are somewhat bulky and weird looking, people playing with a VR headset could set up their smart ring to connect so that their motions could be easily read by the headset, as well as likely being more accurate than other types of VR control. Smart rings could also be used for security and identity authorization, by being able to tap it to things or do a certain motion that verifies that it's a certain person wearing it for 2 factor authentication. There are some challenges with smart rings however, such as battery life. Smart rings require a small form factor and therefore can only have so much power in them. They also have some disadvantages of not having a screen attached compared to how a smart watch does making it so that people couldn't easily respond to texts on them. One thing that could be a challenge but has been handled well by manufacturers is both comfort and looks. It needs to be the right weight so it feels natural on people's hands while still having the capabilities to make it worth buying, and also needs to have a simple looking exterior so that it's similar to other types of rings. Overall I don't see these smart rings becoming the biggest market, as there are already so many devices that can do similar things, but I do think that they're interesting and would consider trying one out myself.